HOMEWORK: April & May AI Review

Everything We Learned: GenAI, Prompting, Agents, and LLM Internals

Student Information

Name:

Date:

SECTION 1: Vocabulary Match (20 points)

Match each term with its definition. Write the letter in the box.

A. Small chunks of text (words, word-pieces, punctuation) that an LLM reads

B. A framework for writing good prompts: Role, Task, Context, Format

C. When AI confidently presents false information as fact

D. The cycle an agent follows: Think -> Act -> Observe -> Repeat

E. Using AI to help you write or improve your prompts

F. A setting that controls how creative vs. predictable AI answers are

G. The amount of text the AI can hold in memory at once

H. Functions an agent can call (web search, run code, send email)

I. Hidden instructions that tell the AI how to behave

J. A mechanism that helps the model focus on important earlier tokens

1. Tokens:

2. RTCF:

3. Hallucination:

4. Agent Loop:

5. Meta Prompting:

6. Temperature:

7. Context Window:

8. Tools:

9. System Prompt:

10. Attention:

SECTION 2: True or False (10 points)

Write T (True) or F (False) in each box.

1. The Transformer architecture was introduced in 2017. [ ]
2. A token is always exactly one full word. [ ]
3. An AI agent can plan steps and use tools; a chatbot cannot. [ ]
4. AGI (Artificial General Intelligence) exists today. [ ]
5. Higher temperature makes AI answers more creative but less predictable. [ ]
6. You should share your passwords with AI chatbots. [ ]
7. Chain-of-thought prompting asks the AI to think step by step. [ ]
8. LLMs truly understand language the same way humans do. [ ]
9. RLHF uses human feedback to improve AI outputs. [ ]
10. When an agent uses a tool, it sends a network request. [ ]

SECTION 3: Multiple Choice (15 points)

Write the letter (A, B, C, or D) in the answer box.

1. What is the core process LLMs use to generate text?

A) Database lookup B) Next-token prediction C) Internet search D) Random selection

Answer: [ ]

2. What does the 'R' in RTCF stand for?

A) Read B) Role C) Run D) Repeat

Answer: [ ]

3. Which task NEEDS an AI agent, not just a chatbot?

A) Define 'happy' B) Tell a joke C) Search, compare, and email a summary D) Solve 2+2

Answer: [ ]

SECTION 3: Multiple Choice (continued)

4. In the agent loop, what comes after 'Act'?

A) Think B) Observe C) Loop D) Goal

Answer:

5. What is 'attention' in a Transformer?

A) AI liking your question B) Focusing on important earlier tokens C) A timer D) Watching you

Answer:

SECTION 4: Short Answer (20 points)

Answer each question in 2-3 sentences.

1. Explain in your own words how an LLM generates text.

Use the words 'token', 'predict', and 'context' in your answer.

2. What is the difference between an AI chatbot and an AI agent?

Give an example task for each.

3. Explain what meta prompting is and why it is useful.

4. Why can LLMs hallucinate? What should you do before trusting an important AI answer?

SECTION 5: Applied Scenarios (25 points)

Real-World Thinking

Read each scenario and apply what you have learned about AI, prompting, agents, and safety to answer the questions.

Scenario A:

Your friend wants to use AI to write a history report about the moon landing.

They type: 'Tell me about the moon.' The AI gives a long, unfocused answer.

Rewrite their prompt using RTCF to get a better result:

Scenario B:

A student asks ChatGPT: 'Who won the NBA finals last night?'

The AI gives a confident answer, but the finals haven't happened yet.

What went wrong? What AI concept explains this? (2-3 sentences)

Scenario C:

You want an AI to: research the 3 best laptops under $500, compare them,

and email a summary to your parent.

Does this need a chatbot or an agent? List the agent components needed:

Scenario D:

You are designing an AI study buddy. Would you set temperature to low or high

for math help? What about for brainstorming creative story ideas? Why?

SECTION 6: Reflection (10 points)

1. What is the most important thing you learned about AI this semester?

2. Name one way AI could help you in your daily life, and one risk
you would need to watch out for.

3. If you could build any AI agent, what would it do? Describe its
tools, system prompt, and one step of its Think->Act->Observe loop.

BONUS: AI Concept Map (+10 points)

Creative Challenge

On a separate sheet (or in the box below), draw or describe a concept map connecting at least 8 of these terms: LLM, Transformer, tokens, next-token prediction, attention, temperature, hallucination, RTCF, meta prompting, AI agent, tools, memory, Think/Act/Observe, context window, system prompt. Show how the concepts relate to each other.

Describe your concept map or attach a photo of your drawing: